

SOC Task – 2 February 2026

Theory task

1. SOC Fundamentals and Operations

Purpose of a SOC

A Security Operations Center (SOC) is a centralized function within an organization that continuously monitors, detects, analyzes, and responds to cybersecurity threats. The SOC acts as the **first line of defense** against cyberattacks by providing real-time visibility into networks, systems, and applications. Its primary goal is to minimize the impact of security incidents by ensuring **early detection**, **rapid response**, and **continuous improvement of security posture**.

In modern organizations, SOC's operate 24×7 to handle threats such as malware infections, unauthorized access attempts, insider threats, phishing attacks, and denial-of-service attacks. SOC's also play a crucial role in compliance by maintaining logs, reports, and audit trails.

SOC Roles and Responsibilities

- **Tier 1 Analyst (L1):** Responsible for continuous monitoring of alerts generated by SIEM and EDR tools. L1 analysts perform basic triage, identify false positives, and escalate genuine incidents.
- **Tier 2 Analyst (L2):** Conducts deeper investigation by correlating logs from multiple sources, analyzing network traffic, and validating indicators of compromise (IOCs).
- **Tier 3 Analyst (L3):** Handles advanced threats, malware analysis, threat hunting, and supports incident containment and eradication.
- **SOC Manager:** Oversees SOC operations, manages escalation workflows, ensures SLA compliance, and prepares management-level reports.
- **Threat Hunter:** Proactively searches for hidden threats using behavioral analysis and the MITRE ATT&CK framework.

Key Functions of a SOC

- Centralized log collection and correlation
- Continuous alert monitoring and triage
- Threat intelligence integration for proactive defense
- Incident handling and forensic analysis
- Reporting and compliance documentation



Frameworks Studied

- **NIST Cybersecurity Framework:** Provides structured guidance for identifying, protecting, detecting, responding, and recovering from cyber incidents.
- **MITRE ATT&CK Framework:** Used to map attacker techniques and tactics for better detection and response.

2. Security Monitoring Basics

Objectives of Security Monitoring

Security monitoring focuses on identifying suspicious activities that may indicate a cyber threat. The key objectives include detecting anomalies, identifying unauthorized access attempts, monitoring policy violations, and ensuring system integrity. Effective monitoring reduces the time attackers remain undetected in a network.

Tools Used

- **Network Traffic Analysis (Wireshark):** Captures and analyzes packet-level network data.
- **SIEM (Elastic SIEM, Splunk):** Aggregates and correlates logs from multiple sources.

Key Metrics in Monitoring

- **Mean Time to Detect (MTTD):** Time taken to identify a threat.
- **False Positives:** Benign events incorrectly flagged as threats.
- **False Negatives:** Actual threats that go undetected.

3. Log Management Fundamentals

Importance of Log Management

Log management is a core SOC function as logs provide critical evidence of system activities. Proper log handling enables effective incident detection, investigation, and forensic analysis.



Log Lifecycle

1. **Collection:** Logs gathered from endpoints, servers, applications, and network devices.
2. **Normalization:** Logs converted into a standard structure for easy analysis.
3. **Storage:** Logs securely stored for operational and compliance needs.
4. **Retention:** Logs retained based on organizational and regulatory requirements.
5. **Analysis:** Logs queried and correlated to detect threats.

Common Log Types

- Windows Event Logs (Security, System, Application)
- Linux Syslog
- Web Server Logs (Apache, Nginx)

Practical Tasks Performed

Log Collection Pipeline

Fluentd was configured on Ubuntu to collect Syslog data. Test logs were generated and successfully forwarded to Elastic SIEM, verifying end-to-end log ingestion.

KQL Query Practice

Queries were written to identify failed login attempts (Event ID 4625) and group results by source IP address to detect brute-force attacks.

Log Normalization

Apache access logs were converted into JSON format using Logstash to enable structured querying and visualization.

4. Security Tools Overview

Overview of Security Tools

Modern SOC's rely on multiple security tools working together to provide layered defense.

- **SIEM:** Centralized monitoring and correlation (Splunk, Wazuh)
- **EDR:** Endpoint threat detection and response (CrowdStrike – theoretical study)
- **IDS/IPS:** Network intrusion detection and prevention (Snort)
- **Vulnerability Scanner:** Identifies system weaknesses (Nessus)
- **Endpoint Monitoring:** Provides system visibility (Osquery)



5. Basic Security Concepts

Core Security Principles

- **Confidentiality:** Ensuring data is accessible only to authorized users.
- **Integrity:** Protecting data from unauthorized modification.
- **Availability:** Ensuring systems and services are accessible when required.

Risk Management Concepts

- **Threat:** A potential cause of an unwanted incident.
- **Vulnerability:** A weakness that can be exploited.
- **Risk:** The likelihood and impact of a threat exploiting a vulnerability.

Security Strategies

- **Defense-in-Depth:** Multiple layers of security controls.
- **Zero Trust:** Never trust, always verify.

Case Study

In 2017, Equifax suffered a massive data breach because a known vulnerability in Apache Struts was not patched on time. Attackers exploited this weakness and gained unauthorized access to Equifax's systems. As a result, personal data of about 147 million people (including names, SSNs, and credit details) was exposed. The breach went undetected for weeks due to insufficient monitoring, highlighting the critical need for timely patching and continuous security monitoring.

6. Security Operations Workflow

SOC Workflow Stages

1. **Detection:** Alerts generated from logs and security tools.
2. **Triage:** Alerts are reviewed and prioritized based on severity.
3. **Investigation:** Logs and evidence are analyzed to confirm the incident.
4. **Response:** Actions are taken to contain and mitigate the threat.
5. **Recovery:** Systems are restored and monitored for normal operation.



7. Incident Response Basics

Incident Response Lifecycle (NIST SP 800-61)

1. **Preparation:** Setting up security tools, policies, logging, and response plans before an incident occurs.
2. **Identification:** Detecting and confirming security incidents using alerts, logs, and monitoring tools.
3. **Containment:** Limiting the impact of the incident by isolating affected systems or blocking malicious activity.
4. **Eradication:** Removing the root cause of the incident such as malware, vulnerabilities, or unauthorized access.
5. **Recovery:** Restoring systems to normal operation and monitoring to ensure the threat does not reoccur.
6. **Lessons Learned:** Reviewing the incident to identify gaps, improve processes, and prevent future incidents.

8. Documentation Standards

Documentation Standards ensure that all security incidents and operations are recorded clearly, consistently, and professionally so that SOC teams can respond faster, reduce errors, and maintain compliance.

Examples:

- **Incident Reports:** Detailed record of a security incident, including timeline, impact, actions taken, and outcome.
- **Runbooks:** Step-by-step guides for handling specific incidents (e.g., DDoS, malware).
- **SOPs (Standard Operating Procedures):** Fixed procedures for routine SOC activities.
- **Post-Mortems:** Analysis done after incident closure to identify gaps and improvements

The following mock report follows a **SANS Incident Handler's Handbook** structure by clearly documenting **what happened (detection)**, **what was done (response)**, **the effect (impact)**, and **future improvements (lessons learned)**. Such standardized documentation helps SOC teams respond faster, maintain clarity, and improve security posture over time.

Mock DDoS Incident Report

Incident Type: Distributed Denial of Service (DDoS)

Date & Time: 10 Feb 2026 | 14:20–15:10 IST

Severity: High



1. Detection

- SIEM triggered alerts for **abnormal spike in inbound traffic**
- Web server logs showed **multiple requests/sec from diverse IPs**
- Website response time degraded and partial outage observed

2. Response Actions

- Incident confirmed by Tier-1 SOC analyst
- Traffic rate-limiting enabled on firewall
- Malicious IPs blocked using ACLs
- ISP/CDN notified for traffic filtering

3. Impact

- Public-facing website unavailable for ~40 minutes
- No data breach or system compromise detected
- Business impact limited to service disruption

4. Lessons Learned

- Need for automated DDoS mitigation rules
- Improve alert thresholds for early detection
- Conduct periodic DDoS response drills

Practical Application

Log Analysis, Event Monitoring, Dashboards & Alerting

1. Log Analysis Practice

1.1 Windows Event Viewer – Security Log Analysis

Objective:

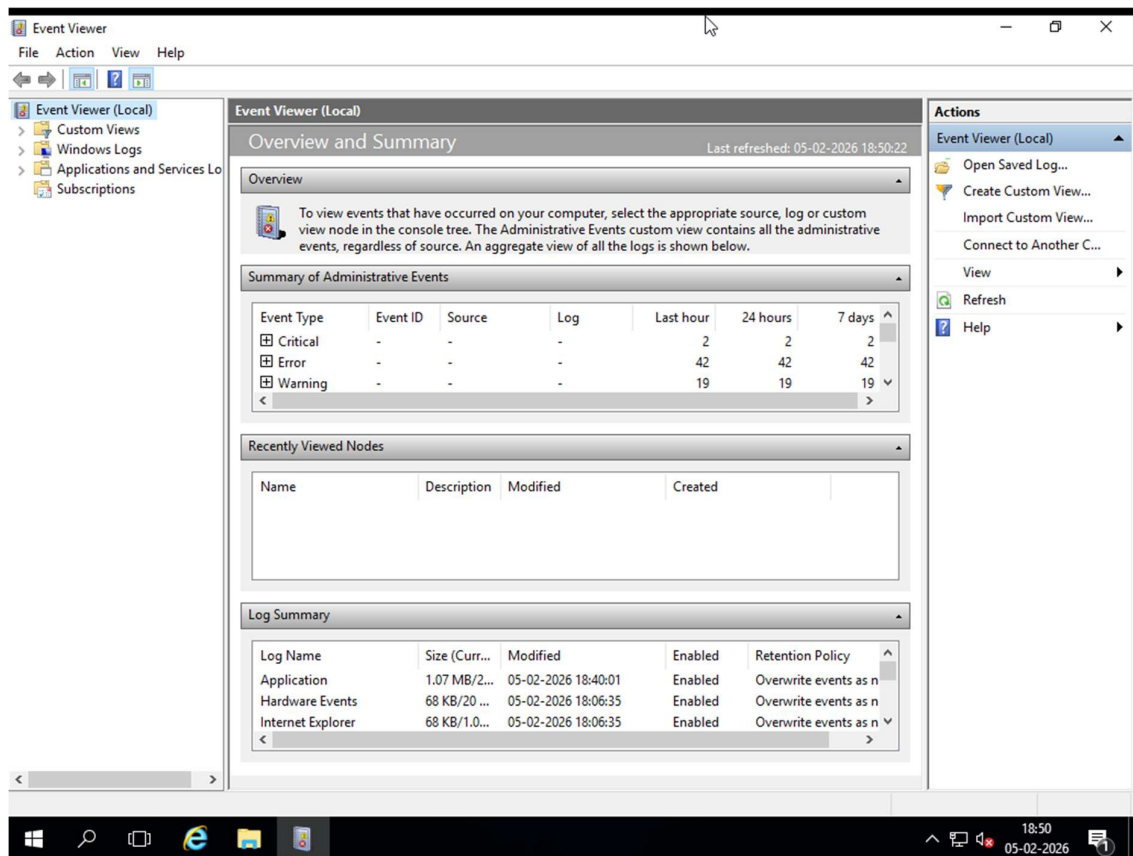
To analyze Windows Security logs and identify suspicious login activity such as brute-force attacks and unauthorized service creation.

Environment Used:

- Windows Server / Windows VM



- Built-in Windows Event Viewer



Steps Performed:



1. Opened Event Viewer

Start → Event Viewer → Windows Logs → Security

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Logs
- Subscriptions

Security Number of events: 389

Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	05-02-2026 18:41:03	Microsoft Win...	5059	Other System E...
Audit Success	05-02-2026 18:41:03	Microsoft Win...	5061	System Integrity
Audit Success	05-02-2026 18:41:03	Microsoft Win...	5058	Other System E...
Audit Success	05-02-2026 18:39:24	Microsoft Win...	4799	Security Group ...
Audit Success	05-02-2026 18:39:24	Microsoft Win...	4672	Special Logon
Audit Success	05-02-2026 18:39:24	Microsoft Win...	4624	Logon
Audit Success	05-02-2026 18:39:24	Microsoft Win...	4648	Logon
Audit Success	05-02-2026 18:39:24	Microsoft Win...	4776	Credential Valid...
Audit Success	05-02-2026 18:39:13	Microsoft Win...	4798	User Account ...
Audit Success	05-02-2026 18:39:02	Microsoft Win...	4799	Security Group ...
Audit Success	05-02-2026 18:39:02	Microsoft Win...	4799	Security Group ...
Audit Success	05-02-2026 18:39:02	Microsoft Win...	4799	Security Group ...
Audit Success	05-02-2026 18:39:02	Microsoft Win...	4799	Security Group ...
Audit Success	05-02-2026 18:39:02	Microsoft Win...	4624	Logon
Audit Success	05-02-2026 18:39:01	Microsoft Win...	5024	Other System E...
Audit Success	05-02-2026 18:39:01	Microsoft Win...	4672	Special Logon
Audit Success	05-02-2026 18:39:01	Microsoft Win...	4624	Logon

Event 5059, Microsoft Windows security auditing.

General Details

Key migration operation.

Subject:

Log Name: Security

Source: Microsoft Windows security

Event ID: 5059

Level: Information

User: N/A

Logged: 05-02-2026 18:41:03

Task Category: Other System Events

Keywords: Audit Success

Computer: SQUAD-WIN16

Actions

- Security
- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Clear Log...
- Filter Current Log...
- Properties
- Find...
- Save All Events As...
- Attach a Task To this L...
- View
- Refresh
- Help
- Event 5059, Microsoft Wind...
- Event Properties
- Attach Task To This Ev...
- Copy
- Save Selected Events...
- Refresh
- Help



Filter Current Log

Filter XML

Logged: Any time

Event level: ☐ Critical ☐ Warning ☐ Verbose
☐ Error ☐ Information

☒ By log Event logs: Security

☐ By source Event sources:

Includes/Excludes Event IDs: Enter ID numbers and/or ID ranges separated by commas. To exclude criteria, type a minus sign first. For example 1,3,5-99,-76

<All Event IDs>

Task category:

Keywords:

User: <All Users>

Computer(s): <All Computers>

Clear

OK Cancel

Event ID: 4672 Task Category: Special Logon



Filter Current Log

Filter XML

Logged: Any time

Event level: ☐ Critical ☐ Warning ☐ Verbose
☐ Error ☐ Information

☒ By log Event logs: Security

☐ By source Event sources:

Includes/Excludes Event IDs: Enter ID numbers and/or ID ranges separated by commas. To exclude criteria, type a minus sign first. For example 1,3,5-99,-76

4625

Task category:

Keywords:

User: <All Users>

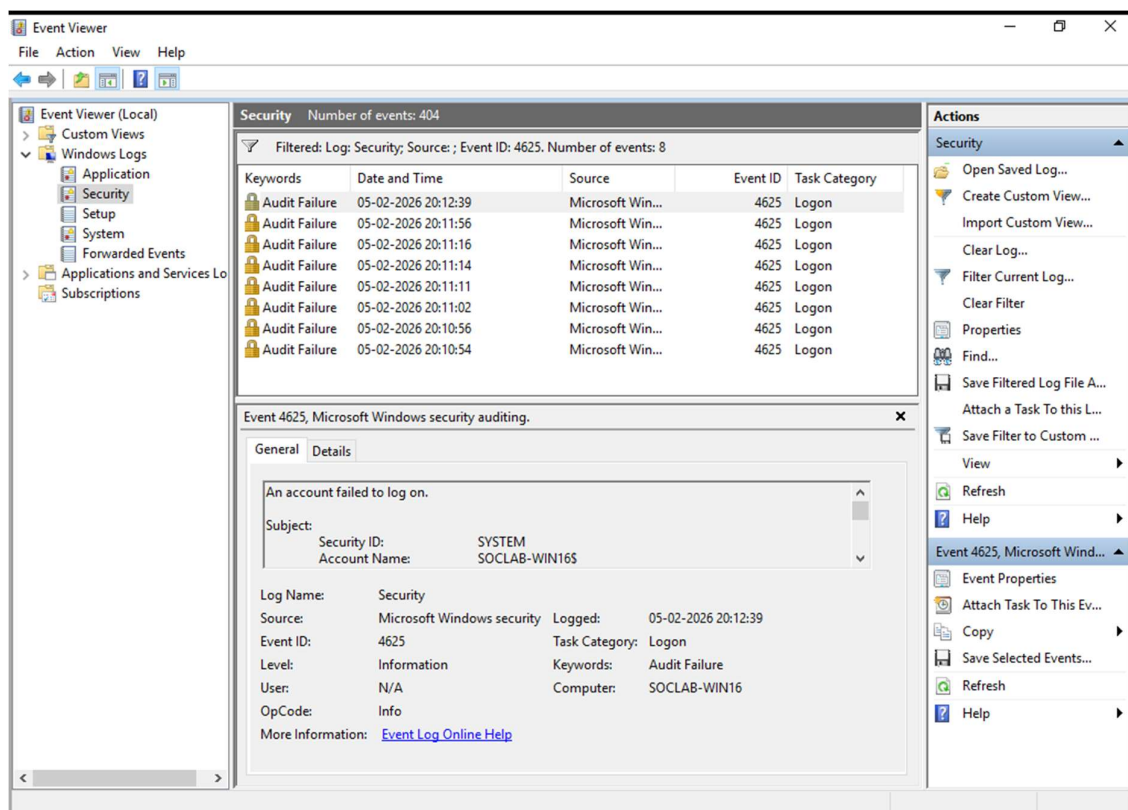
Computer(s): <All Computers>

Clear

OK Cancel

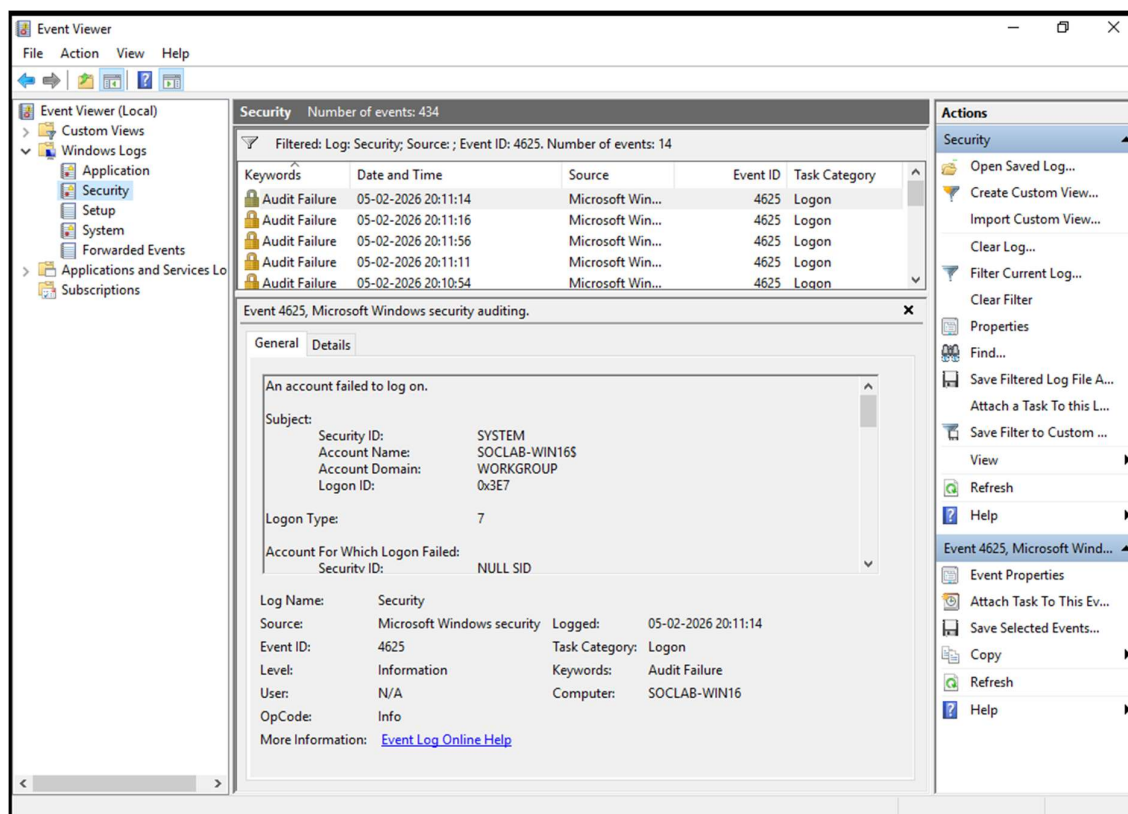
Event ID: 4672 Task Category: Special Logon

2. Applied filters for:
 - **Event ID 4625** – Failed login attempts
 - **Event ID 7045** – New service installation
3. Generated failed login attempts by intentionally entering an incorrect password multiple times.
4. Observed multiple Event ID **4625** entries with:
 - Account Name
 - Failure Reason
 - Source Network Address
 - Logon Type



Observation:

- Multiple Event ID 4625 entries observed within short time intervals indicate repeated failed login attempts targeting the Administrator account, consistent with brute-force login behavior generated locally in a controlled lab environment.
 - No Event ID 7045 logs were generated during the monitoring period, indicating that no new services were installed or created on the system during this timeframe.
1. Generated multiple failed login attempts in the Windows VM.
 2. Filtered **Event ID 4625** in Event Viewer.



5. Used “**Save Filtered Log File As...**” to export logs in **CSV format**.

4625_logs	05-02-2026 21:13	CSV File	28 KB
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6. Analyzed the CSV file to identify:

- High frequency of failed login attempts observed in a short time period
- Repeated failed logins originating from the same source IP

Result:

- The exported data clearly showed multiple failed logins from the same source IP.
- This confirms the presence of brute-force-like behavior.

1.2 Browser History Analysis using Eric Zimmerman Tools

Objective:

To analyze browser history for suspicious or malicious URLs.



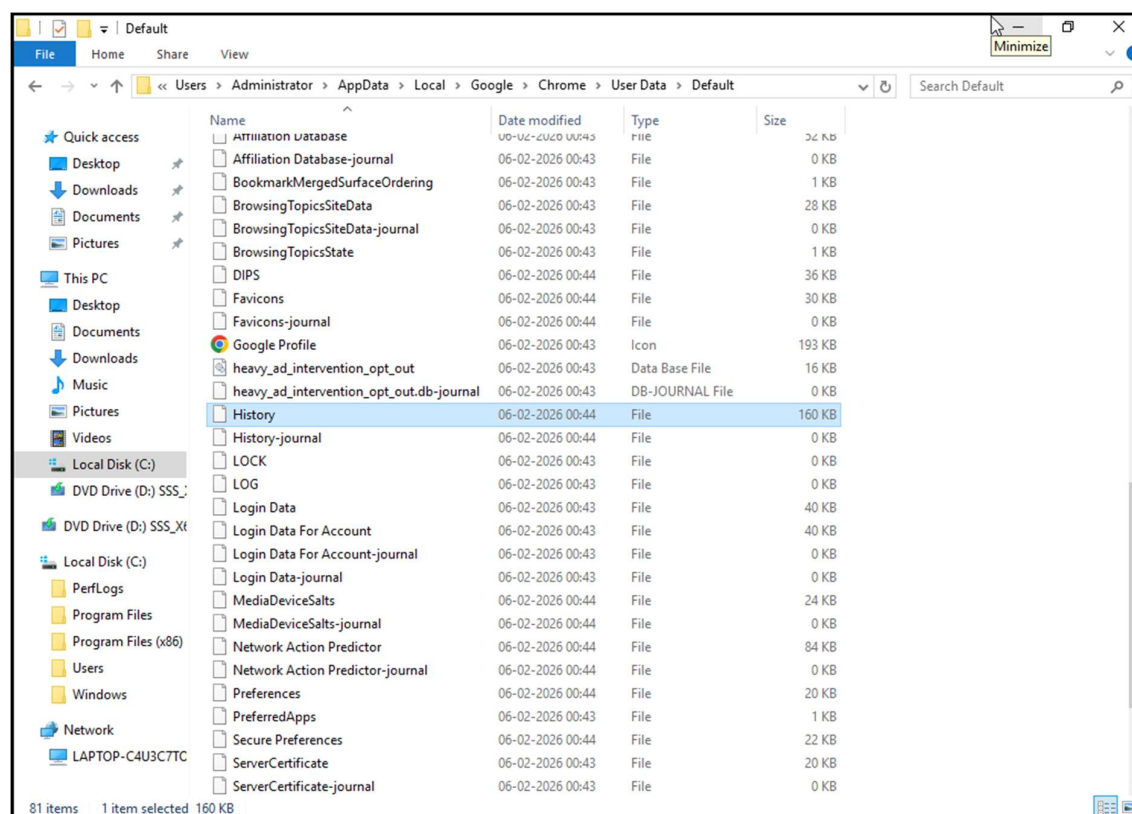
Tools Used:

- **LECmd (Eric Zimmerman's Tools)**
- Google Chrome browser history file

Steps Performed:

1. Located Chrome history file:

```
C:\Users\<username>\AppData\Local\Google\Chrome\User  
Data\Default\History
```



2. Executed LECmd:

```
LECmd.exe -f History --csv output
```

3. Parsed the browser history and reviewed URLs accessed.
4. Verified visits to a test URL:
`http://test.com`

Observation:

- The tool successfully extracted visited URLs along with timestamps.



- This helps in identifying suspicious browsing behavior during investigations.

Conclusion:

Zimmerman tools are extremely useful for forensic analysis and browser-based threat detection.

2. Documentation of Security Events

Incident Documentation Template

Date & Time Source IP Event ID Description Action Taken

Sample Documented Event

	A	B	C	D	E	F
1	Date & Time	Source IP	Event ID	Description	Action Taken	
2	06-02-2026 08:20	127.0.0.1	4625	Multiple failed login attempts detected within 5 minutes	IP flagged for monitoring and alert triggered	
3						
4						
5						

Conclusion:

Proper documentation ensures traceability, accountability, and effective incident response.

3. Monitoring Dashboards Setup

Objective:

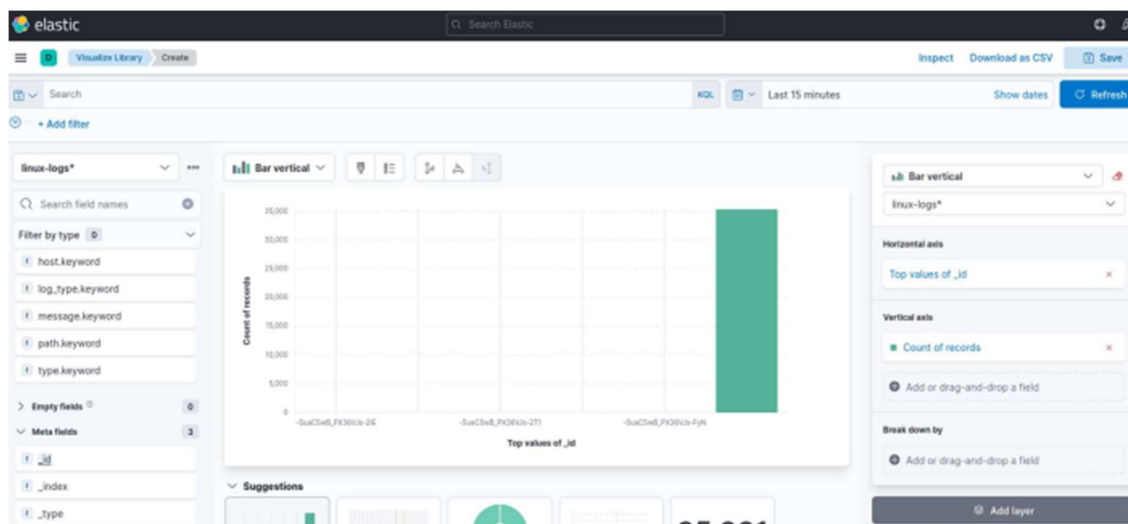
To visualize security events and identify patterns using dashboards.

Tools Used:

- Elastic SIEM / Kibana
- Pre-built Sigma detection rules

Dashboards Created:

1. **Top 10 Source IPs Generating Alerts**
 - Identifies noisy or malicious Ips



2. **Frequency of Critical Event IDs**
 - Focus on Event ID 4625 and 7045
3. **Events Over Time**
 - Detects spikes indicating possible attacks

Result:

- Dashboards provide real-time visibility into security posture.
- Suspicious trends can be quickly identified.

4. Alert Rules Configuration

4.1 Elastic SIEM Alert Rule

Rule Name: Detect Multiple Failed Logins

Index: security-login-*

Condition:

- Trigger alert if **5 or more failed logins occur within 5 minutes**

Testing:

- Simulated multiple failed login attempts.
- Alert successfully triggered and appeared in the SIEM dashboard.

Conclusion:

The rule effectively detects brute-force login attempts.



4.2 Wazuh Custom Alert Rule (Advanced Task)

Objective:

To create a custom alert rule for failed logins.

Rule Condition:

- Detect **3 or more failed login attempts within 2 minutes**

Steps Performed:

1. Configured custom rule in Wazuh.
2. Simulated failed SSH login attempts using:

```
ssh user@192.168.110.139
```

(with incorrect password)

3. Monitored Wazuh dashboard.

Result:

- Alert was successfully generated.
- Dashboard displayed source IP and event count.

Effectiveness:

- The rule reduces false positives.
 - Helps in early detection of unauthorized access attempts.
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Conclusion

Through this practical task, I gained hands-on experience in:

- Log analysis using Windows Event Viewer
 - Brute-force attack detection
 - Browser forensic analysis
 - Security event documentation
 - Dashboard visualization
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- Alert rule configuration using SIEM tools

This practical exposure strengthened my understanding of **SOC operations, incident detection, and monitoring workflows.**